



T.C.

**ADANA ALPARSLAN TÜRKEŞ BİLİM VE
TEKNOLOJİ ÜNİVERSİTESİ**

**BİLGİSAYAR VE BİLİŞİM FAKÜLTESİ
BİLGİSAYAR MÜHENDİSLİĞİ**

**AI PROJECT – TEXT SUMMARIZATION
REPORT**

Öğrenci Adı ve Soyadı:KADİR CAN GÖNÜLLÜ

DERYA LARA KESTİ

EMRE YELOĞLU

Öğrenci Numarası : 210101121

210101130

210101125

Report: Document and Dataset Summarization App

Project Overview

This project is designed to develop an advanced application for text summarization. Leveraging natural language processing (NLP) techniques, the application summarizes both user-uploaded documents (PDFs) and articles from datasets. Additionally, it evaluates the model's summarization performance using ROUGE scores.

Technical Detailed Analysis

Libraries and Their Usage

1. Streamlit

- a. Provides a user interface for creating a web-based application.
- b. Features such as custom tab structures, file upload, and data visualization are implemented.

2. Transformers

- a. Hugging Face's Transformers library is used to load the language model (T5) and tokenizer.
- b. The **pipeline** structure simplifies the summarization process.

3. PyPDF2

- a. Reads PDF files and extracts text page by page.

4. Pandas

- a. Handles dataset loading, processing, and visualization for the user.

5. ROUGE Scorer

- a. Computes ROUGE scores to compare the model-generated summary with the reference summary.

6. PyTorch

- a. Handles model weights and optimizes performance on GPU/CPU.

7. Base64

- a. Encodes PDF content for embedding in an HTML format.

General Project Structure

Document and Dataset Summarization App ^{GO}

Summarize PDF/Custom Text Test Dataset

Upload your PDF File



Drag and drop file here
Limit 200MB per file • PDF

Browse files

Or enter your text below:

Summarize PDF Text

Summarize Custom Text

❖ Language Model and Tokenizer

- The project uses the **LaMini-Flan-T5-248M** model.
- The T5Tokenizer converts input text into a format understandable by the model.
- The T5ForConditionalGeneration class performs the summarization task.

Customization Parameters:

- `max_length=500`: Maximum summary length.
- `min_length=50`: Minimum summary length.
- `torch_dtype=torch.float32`: Uses 32-bit floating-point operations for model processing.

❖ PDF Processing

- PyPDF2 extracts text from uploaded PDF files, page by page.
- The extracted text is displayed and summarized upon user request.

❖ Text Summarization

- The pipeline function performs the summarization task.
- Input text is passed to the model, and the summarized output is returned and displayed to the user.

Summarize PDF/Custom Text Test Dataset

Dataset Summarization and Evaluation

Enter the path to your dataset (CSV):

Please provide the path to the dataset.

❖ ROUGE Score Calculation

- ROUGE1, ROUGE2, and ROUGE-L metrics are used to evaluate model performance.
- The `rouge_scorer.RougeScorer` class compares the reference summary with the model-generated summary.

ROUGE Scores:

```
{
  "rouge1": [
    0 : 0.26136363636363635
    1 : 0.6764705882352942
    2 : 0.3770491803278689
  ]
  "rouge2": [
    0 : 0.19540229885057472
    1 : 0.5151515151515151
    2 : 0.28333333333333334
  ]
  "rougeL": [
    0 : 0.17045454545454544
    1 : 0.4411764705882353
    2 : 0.2459016393442623
  ]
}
```

User Interface Features

1. Tab Structure

- The interface has two tabs:
 - Summarize PDF/Custom Text:** Performs summarization on PDF files or user-provided text.
 - Test Dataset:** Tests the model on a dataset and displays ROUGE scores.

2. PDF Upload and Summarization

- Users can upload a PDF file.
- Text is extracted, summarized, and displayed.

3. Custom Text Summarization

- a. Users can manually input text and generate a summary.

4. Dataset Testing

- a. Users provide a CSV file containing article and highlights columns.
- b. For each row:
 - i. The article column is summarized.
 - ii. The highlights column (reference summary) is compared to the generated summary.
 - iii. ROUGE scores are calculated and displayed.

First few rows of the dataset:

	id	article	highlights
0	0001d1afc246a7964130f43ae940af6b6c57f01	By .Associated Press . PUBLISHED: . 14:11 EST, 25 October 2013 . . UPDATED: . 15:36	Bishop John Folda, of North Dakota, is taking time off after being diagnosed . He cont
1	0002095e55fcb3a2f966d98f92a95433dc305ef	(CNN) -- Ralph Mata was an internal affairs lieutenant for the Miami-Dade Police Depa	Criminal complaint: Cop used his role to help cocaine traffickers . Ralph Mata, an inte
2	00027e965c8264c35cc1bc55556db388da82b07f	A drunk driver who killed a young woman in a head-on crash while checking his mob	Craig Eccleston-Todd, 27, had drunk at least three pints before driving car . Was using
3	0002c17436637c4fe1837c935c0ade47adb18e9a	(CNN) -- With a breezy sweep of his pen President Vladimir Putin wrote a new chapter	Nina dos Santos says Europe must be ready to accept sanctions will hurt both sides .
4	0003ad6ef9c37534f80b55b4235108024b4b07f0b	Fleetwood are the only team still to have a 100% record in Sky Bet League One as a 2	Fleetwood top of League One after 2-0 win at Scunthorpe . Peterborough, Bristol City

Select a row to summarize

0

Original Article:

By .Associated Press . PUBLISHED: . 14:11 EST, 25 October 2013 . | . UPDATED: . 15:36 EST, 25 October 2013 . The bishop of the Fargo Catholic Diocese in North Dakota has exposed potentially hundreds of church members in Fargo, Grand Forks and Jamestown to the hepatitis A virus in late September and early October. The state Health Department has issued an advisory of exposure for anyone who attended five churches and took communion. Bishop John Folda (pictured) of the Fargo Catholic Diocese in North Dakota has exposed potentially hundreds of church members in Fargo, Grand Forks and Jamestown to the hepatitis A . State Immunization Program Manager Molly Howell says the risk is low, but officials feel it's important to alert people to the possible exposure. The diocese announced on Monday that Bishop John Folda is taking time off after being diagnosed with hepatitis A. The diocese says he contracted the infection through contaminated food while attending a conference for newly ordained bishops in Italy last month. Symptoms of hepatitis A include fever, tiredness, loss of appetite, nausea and abdominal discomfort. Fargo Catholic Diocese in North Dakota (pictured) is where the bishop is located .

True Summary:

Bishop John Folda, of North Dakota, is taking time off after being diagnosed .
He contracted the infection through contaminated food in Italy .
Church members in Fargo, Grand Forks and Jamestown could have been exposed .

Predicted Summary:

Bishop John Folda of the Fargo Catholic Diocese in North Dakota has exposed potentially hundreds of church members in Fargo, Grand Forks, and Jamestown to the hepatitis A virus in late September and early October. The state Health Department has issued an advisory of exposure for anyone who attended five churches and took communion. Bishop Folda contracted the infection through contaminated food while attending a conference for newly ordained bishops in Italy last month. Symptoms of the virus include fever, tiredness, loss of appetite, nausea, and abdominal discomfort.

Critical Code Components

1. Summarization Function

```
def summarize_text(input_text):  
    result = pipe_sum(input_text)  
    return result[0]['summary_text']
```

This function uses the pipeline to summarize the provided text.

2. ROUGE Score Calculation

```
def calculate_rouge(true_summary, predicted_summary):
    scorer = rouge_scorer.RougeScorer(['rouge1', 'rouge2',
    'rougeL'], use_stemmer=True)
    scores = scorer.score(true_summary, predicted_summary)
    return scores
```

Calculates evaluation metrics to assess summarization performance.

3. Dataset Processing

```
data = pd.read_csv(dataset_path)
if 'article' not in data.columns or 'highlights' not in
data.columns:
    st.error("Dataset must contain 'article' and 'highlights'
columns.")
else:
    # Process dataset
```

Ensures the provided dataset is in the correct format and processes it accordingly.

Application Use Cases

1. Education

- a. Summarizing lecture notes for quick review.

2. Research

- a. Generating concise summaries of academic papers.

3. Business

- a. Producing brief summaries of reports and documents.

4. Media

- a. Creating short, informative summaries of news articles.

Conclusion

This project effectively leverages language models and text processing techniques to generate summaries. Its user-friendly interface and ROUGE evaluation feature make it a powerful tool for education, research, and business. The app demonstrates robust capabilities in summarizing text and evaluating performance, providing practical value across multiple domains.